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D:\PycharmProject\Deep-Reinforcement-Learning-Hands-On  
\.venv\Scripts\python.exe D:\PycharmProject\Deep-  
Reinforcement-Learning-Hands-On\Chapter_06\  
01_DQN_on_Pong.py
```

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A.L.E: Arcade Learning Environment (version 0.12.1+  
8a8fafb)
```

```
[Powered by Stella]
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```
Episode: 000001; Score: -21; Epsilon: 0.985  
Episode: 000002; Score: -19; Epsilon: 0.979  
Episode: 000003; Score: -19; Epsilon: 0.974  
Episode: 000004; Score: -21; Epsilon: 0.970  
Episode: 000005; Score: -21; Epsilon: 0.966  
Episode: 000006; Score: -20; Epsilon: 0.961  
Episode: 000007; Score: -21; Epsilon: 0.957  
Episode: 000008; Score: -20; Epsilon: 0.953  
Episode: 000009; Score: -21; Epsilon: 0.949  
Episode: 000010; Score: -20; Epsilon: 0.944  
Episode: 000011; Score: -19; Epsilon: 0.939  
Episode: 000012; Score: -20; Epsilon: 0.934  
Episode: 000013; Score: -20; Epsilon: 0.929  
Episode: 000014; Score: -21; Epsilon: 0.924  
Episode: 000015; Score: -20; Epsilon: 0.920  
Episode: 000016; Score: -21; Epsilon: 0.915  
Episode: 000017; Score: -21; Epsilon: 0.911  
Episode: 000018; Score: -21; Epsilon: 0.907  
Episode: 000019; Score: -20; Epsilon: 0.902  
Episode: 000020; Score: -21; Epsilon: 0.898  
Episode: 000021; Score: -20; Epsilon: 0.893  
Episode: 000022; Score: -21; Epsilon: 0.888  
Episode: 000023; Score: -21; Epsilon: 0.884  
Episode: 000024; Score: -20; Epsilon: 0.879  
Episode: 000025; Score: -20; Epsilon: 0.875  
Episode: 000026; Score: -21; Epsilon: 0.870  
Episode: 000027; Score: -21; Epsilon: 0.866  
Episode: 000028; Score: -21; Epsilon: 0.862  
Episode: 000029; Score: -21; Epsilon: 0.858  
Episode: 000030; Score: -20; Epsilon: 0.853  
Episode: 000031; Score: -20; Epsilon: 0.848  
Episode: 000032; Score: -21; Epsilon: 0.844  
Episode: 000033; Score: -21; Epsilon: 0.839  
Episode: 000034; Score: -20; Epsilon: 0.834
```

Episode: 000035;	Score: -21;	Epsilon: 0.830
Episode: 000036;	Score: -19;	Epsilon: 0.824
Episode: 000037;	Score: -20;	Epsilon: 0.820
Episode: 000038;	Score: -21;	Epsilon: 0.815
Episode: 000039;	Score: -21;	Epsilon: 0.811
Episode: 000040;	Score: -20;	Epsilon: 0.806
Episode: 000041;	Score: -20;	Epsilon: 0.801
Episode: 000042;	Score: -18;	Epsilon: 0.795
Episode: 000043;	Score: -20;	Epsilon: 0.790
Episode: 000044;	Score: -19;	Epsilon: 0.784
Episode: 000045;	Score: -21;	Epsilon: 0.778
Episode: 000046;	Score: -18;	Epsilon: 0.773
Episode: 000047;	Score: -20;	Epsilon: 0.768
Episode: 000048;	Score: -19;	Epsilon: 0.761
Episode: 000049;	Score: -18;	Epsilon: 0.755
Episode: 000050;	Score: -21;	Epsilon: 0.751
Episode: 000051;	Score: -21;	Epsilon: 0.746
Episode: 000052;	Score: -17;	Epsilon: 0.739
Episode: 000053;	Score: -19;	Epsilon: 0.732
Episode: 000054;	Score: -20;	Epsilon: 0.727
Episode: 000055;	Score: -18;	Epsilon: 0.722
Episode: 000056;	Score: -16;	Epsilon: 0.714
Episode: 000057;	Score: -18;	Epsilon: 0.707
Episode: 000058;	Score: -17;	Epsilon: 0.701
Episode: 000059;	Score: -18;	Epsilon: 0.694
Episode: 000060;	Score: -17;	Epsilon: 0.688
Episode: 000061;	Score: -19;	Epsilon: 0.683
Episode: 000062;	Score: -20;	Epsilon: 0.678
Episode: 000063;	Score: -17;	Epsilon: 0.671
Episode: 000064;	Score: -19;	Epsilon: 0.665
Episode: 000065;	Score: -19;	Epsilon: 0.659
Episode: 000066;	Score: -17;	Epsilon: 0.653
Episode: 000067;	Score: -20;	Epsilon: 0.647
Episode: 000068;	Score: -19;	Epsilon: 0.640
Episode: 000069;	Score: -21;	Epsilon: 0.636
Episode: 000070;	Score: -18;	Epsilon: 0.630
Episode: 000071;	Score: -17;	Epsilon: 0.623
Episode: 000072;	Score: -20;	Epsilon: 0.619
Episode: 000073;	Score: -19;	Epsilon: 0.613
Episode: 000074;	Score: -19;	Epsilon: 0.606
Episode: 000075;	Score: -21;	Epsilon: 0.600

Episode: 000076;	Score: -21;	Epsilon: 0.595
Episode: 000077;	Score: -20;	Epsilon: 0.589
Episode: 000078;	Score: -20;	Epsilon: 0.583
Episode: 000079;	Score: -17;	Epsilon: 0.576
Episode: 000080;	Score: -18;	Epsilon: 0.569
Episode: 000081;	Score: -18;	Epsilon: 0.562
Episode: 000082;	Score: -19;	Epsilon: 0.556
Episode: 000083;	Score: -16;	Epsilon: 0.548
Episode: 000084;	Score: -19;	Epsilon: 0.542
Episode: 000085;	Score: -18;	Epsilon: 0.535
Episode: 000086;	Score: -19;	Epsilon: 0.529
Episode: 000087;	Score: -16;	Epsilon: 0.521
Episode: 000088;	Score: -18;	Epsilon: 0.513
Episode: 000089;	Score: -19;	Epsilon: 0.505
Episode: 000090;	Score: -20;	Epsilon: 0.498
Episode: 000091;	Score: -18;	Epsilon: 0.491
Episode: 000092;	Score: -21;	Epsilon: 0.484
Episode: 000093;	Score: -18;	Epsilon: 0.476
Episode: 000094;	Score: -15;	Epsilon: 0.467
Episode: 000095;	Score: -18;	Epsilon: 0.460
Episode: 000096;	Score: -20;	Epsilon: 0.454
Episode: 000097;	Score: -19;	Epsilon: 0.446
Episode: 000098;	Score: -16;	Epsilon: 0.438
Episode: 000099;	Score: -16;	Epsilon: 0.430
Episode: 000100;	Score: -18;	Epsilon: 0.422
Episode: 000101;	Score: -18;	Epsilon: 0.413
Episode: 000102;	Score: -19;	Epsilon: 0.407
Episode: 000103;	Score: -18;	Epsilon: 0.400
Episode: 000104;	Score: -20;	Epsilon: 0.392
Episode: 000105;	Score: -20;	Epsilon: 0.384
Episode: 000106;	Score: -17;	Epsilon: 0.375
Episode: 000107;	Score: -19;	Epsilon: 0.368
Episode: 000108;	Score: -20;	Epsilon: 0.361
Episode: 000109;	Score: -18;	Epsilon: 0.352
Episode: 000110;	Score: -20;	Epsilon: 0.344
Episode: 000111;	Score: -17;	Epsilon: 0.335
Episode: 000112;	Score: -20;	Epsilon: 0.327
Episode: 000113;	Score: -15;	Epsilon: 0.318
Episode: 000114;	Score: -19;	Epsilon: 0.309
Episode: 000115;	Score: -19;	Epsilon: 0.301
Episode: 000116;	Score: -18;	Epsilon: 0.293

Episode: 000117;	Score: -19;	Epsilon: 0.286
Episode: 000118;	Score: -18;	Epsilon: 0.277
Episode: 000119;	Score: -17;	Epsilon: 0.269
Episode: 000120;	Score: -15;	Epsilon: 0.257
Episode: 000121;	Score: -18;	Epsilon: 0.247
Episode: 000122;	Score: -17;	Epsilon: 0.239
Episode: 000123;	Score: -19;	Epsilon: 0.232
Episode: 000124;	Score: -14;	Epsilon: 0.222
Episode: 000125;	Score: -18;	Epsilon: 0.211
Episode: 000126;	Score: -18;	Epsilon: 0.200
Episode: 000127;	Score: -13;	Epsilon: 0.188
Episode: 000128;	Score: -16;	Epsilon: 0.179
Episode: 000129;	Score: -16;	Epsilon: 0.169
Episode: 000130;	Score: -18;	Epsilon: 0.160
Episode: 000131;	Score: -19;	Epsilon: 0.151
Episode: 000132;	Score: -13;	Epsilon: 0.139
Episode: 000133;	Score: -18;	Epsilon: 0.131
Episode: 000134;	Score: -14;	Epsilon: 0.119
Episode: 000135;	Score: -10;	Epsilon: 0.108
Episode: 000136;	Score: -16;	Epsilon: 0.099
Episode: 000137;	Score: -13;	Epsilon: 0.089
Episode: 000138;	Score: -16;	Epsilon: 0.081
Episode: 000139;	Score: -14;	Epsilon: 0.069
Episode: 000140;	Score: -14;	Epsilon: 0.059
Episode: 000141;	Score: -15;	Epsilon: 0.050
Episode: 000142;	Score: -12;	Epsilon: 0.050
Episode: 000143;	Score: -7;	Epsilon: 0.050
Episode: 000144;	Score: -4;	Epsilon: 0.050
Episode: 000145;	Score: -11;	Epsilon: 0.050
Episode: 000146;	Score: 1;	Epsilon: 0.050
Episode: 000147;	Score: -13;	Epsilon: 0.050
Episode: 000148;	Score: -4;	Epsilon: 0.050
Episode: 000149;	Score: -7;	Epsilon: 0.050
Episode: 000150;	Score: -7;	Epsilon: 0.050
Episode: 000151;	Score: -7;	Epsilon: 0.050
Episode: 000152;	Score: -2;	Epsilon: 0.050
Episode: 000153;	Score: -5;	Epsilon: 0.050
Episode: 000154;	Score: 2;	Epsilon: 0.050
Episode: 000155;	Score: -1;	Epsilon: 0.050
Episode: 000156;	Score: -13;	Epsilon: 0.050
Episode: 000157;	Score: 4;	Epsilon: 0.050

Episode: 000158;	Score: 9;	Epsilon: 0.050
Episode: 000159;	Score: 6;	Epsilon: 0.050
Episode: 000160;	Score: 5;	Epsilon: 0.050
Episode: 000161;	Score: 7;	Epsilon: 0.050
Episode: 000162;	Score: 5;	Epsilon: 0.050
Episode: 000163;	Score: -6;	Epsilon: 0.050
Episode: 000164;	Score: 7;	Epsilon: 0.050
Episode: 000165;	Score: 11;	Epsilon: 0.050
Episode: 000166;	Score: 7;	Epsilon: 0.050
Episode: 000167;	Score: 13;	Epsilon: 0.050
Episode: 000168;	Score: 3;	Epsilon: 0.050
Episode: 000169;	Score: 1;	Epsilon: 0.050
Episode: 000170;	Score: 5;	Epsilon: 0.050
Episode: 000171;	Score: 9;	Epsilon: 0.050
Episode: 000172;	Score: 8;	Epsilon: 0.050
Episode: 000173;	Score: 13;	Epsilon: 0.050
Episode: 000174;	Score: 5;	Epsilon: 0.050
Episode: 000175;	Score: 11;	Epsilon: 0.050
Episode: 000176;	Score: 13;	Epsilon: 0.050
Episode: 000177;	Score: 18;	Epsilon: 0.050
Episode: 000178;	Score: 4;	Epsilon: 0.050
Episode: 000179;	Score: 7;	Epsilon: 0.050
Episode: 000180;	Score: 12;	Epsilon: 0.050
Episode: 000181;	Score: 15;	Epsilon: 0.050
Episode: 000182;	Score: 9;	Epsilon: 0.050
Episode: 000183;	Score: 4;	Epsilon: 0.050
Episode: 000184;	Score: 3;	Epsilon: 0.050
Episode: 000185;	Score: 12;	Epsilon: 0.050
Episode: 000186;	Score: -1;	Epsilon: 0.050
Episode: 000187;	Score: 3;	Epsilon: 0.050
Episode: 000188;	Score: 6;	Epsilon: 0.050
Episode: 000189;	Score: 6;	Epsilon: 0.050
Episode: 000190;	Score: 11;	Epsilon: 0.050
Episode: 000191;	Score: 4;	Epsilon: 0.050
Episode: 000192;	Score: 4;	Epsilon: 0.050
Episode: 000193;	Score: 4;	Epsilon: 0.050
Episode: 000194;	Score: 9;	Epsilon: 0.050
Episode: 000195;	Score: 12;	Epsilon: 0.050
Episode: 000196;	Score: 11;	Epsilon: 0.050
Episode: 000197;	Score: 5;	Epsilon: 0.050
Episode: 000198;	Score: 5;	Epsilon: 0.050

```
Episode: 000199; Score: 11; Epsilon: 0.050
Episode: 000200; Score: 11; Epsilon: 0.050
Episode: 000201; Score: 11; Epsilon: 0.050
Episode: 000202; Score: 11; Epsilon: 0.050
Episode: 000203; Score: 11; Epsilon: 0.050
Episode: 000204; Score: 2; Epsilon: 0.050
Episode: 000205; Score: 12; Epsilon: 0.050
Episode: 000206; Score: 17; Epsilon: 0.050
Episode: 000207; Score: 12; Epsilon: 0.050
Episode: 000208; Score: 10; Epsilon: 0.050
Episode: 000209; Score: 10; Epsilon: 0.050
Episode: 000210; Score: -2; Epsilon: 0.050
Episode: 000211; Score: 8; Epsilon: 0.050
Episode: 000212; Score: 7; Epsilon: 0.050
Episode: 000213; Score: 14; Epsilon: 0.050
Episode: 000214; Score: 9; Epsilon: 0.050
Episode: 000215; Score: 14; Epsilon: 0.050
Episode: 000216; Score: 4; Epsilon: 0.050
Episode: 000217; Score: 20; Epsilon: 0.050
Solved!
```

进程结束退出代码为0